

Name: _____, Roll No.: _____

Total marks:

CSE 350/550: Network Security
Quiz 1: Jan 29, 2026

Instructions:

- There are 4 questions, each 7 marks, for a total of 28 marks. Total time available 30 minutes.
- Provide answers only in the space provided. You may do your rough work on a separate blank sheet.
- You may use a calculator, but nothing else. **Pl. keep your phones/laptops, books etc. away from you.**
- Do not use unfair means, else action will be taken. Do not use material such as books, notes, computer, smart phone, etc. And do not share information with other students.

Marks,
Q1:

1. Consider a crypto-system similar to Caesar-cipher, but where $c = p * k \bmod 26$. Here the set of symbols is $\{0, 1, 2, \dots, 24, 25\}$, p is plaintext, c is ciphertext, k is key. For example, if $\text{key}=3$, $p=15$, then $c=19$. Is there a constraint on the value of key, k ? If so, what condition must k satisfy to qualify as a valid key. And give two examples of k that are NOT allowed, and another 2 examples of k that are allowed.

Condition that k must satisfy to qualify as a valid key: **k and 26 should be co-prime. That is, k should NOT have 13 or 2 as a factor** **3 marks**

Give TWO examples of k that are NOT allowed:

i and ii: **k is either 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 or 13** **2 marks**

Give TWO examples of k that are allowed:

i and ii: **k is either 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25** **2 marks**

Marks,
Q2:

2. Consider the poly-alphabetic substitution cipher discussed in class, where the key is 'deceptive'. If the **ciphertext** is
z l j e k x j z l q h k w r h d z v h h
then what is the plaintext? Also show your working in the table below.

Plaintext: **we have been discovered** **3 marks**

key	d	e	c	e	p	t	i	v	e	d	e	c	e	p	t	i	v	e	d	e
numeric key	3	4	2	4	15	19	8	21	4	3	4	2	4	15	19	8	21	4	3	4
plaintext	w	e	h	a	v	e	b	e	e	n	d	i	s	c	o	v	e	r	e	d
ciphertext	z	i	j	e	k	x	j	z	i	q	h	k	w	r	h	d	z	v	h	h

**2 marks for the correct
numeric keys in table above**

**2 marks the correct
plaintext in table above**

Marks,
Q3:

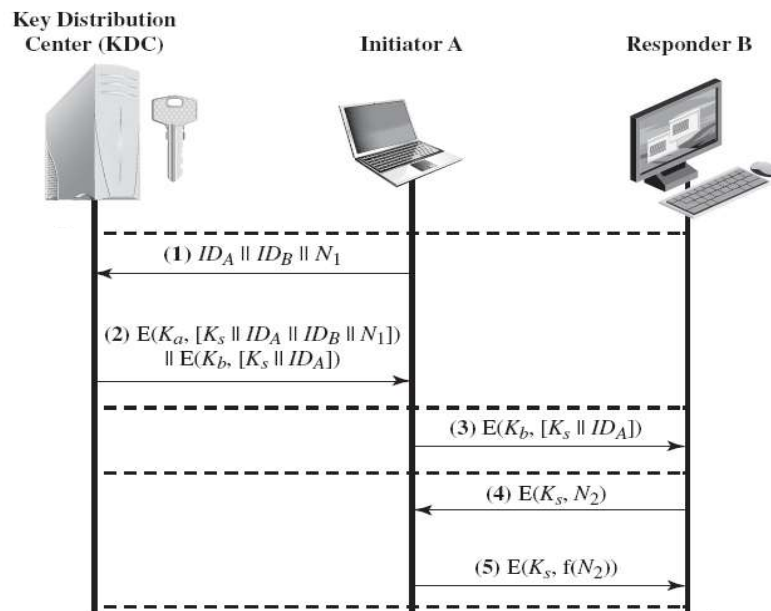
3. The initial permutation (before round 1) in DES is described by an 8x8 matrix. This is given below. The final permutation (after round 16) reverses the permutation carried out in the initial permutation. This is also described by an 8x8 matrix. Only a small portion of this matrix is shown below. It also shows that the 1-st output bit is the 40-th input bit into the final permutation. In what way are the 2-nd, 3-rd, ..., 8-th output bits related to the input bits of the final permutation? Just fill in the blank portions of the table.

<i>Initial Permutation</i>	<i>Final Permutation</i>
58 50 42 34 26 18 10 02	40 08 48 16 56 24 64 3
60 52 44 36 28 20 12 04	
62 54 46 38 30 22 14 06	
64 56 48 40 32 24 16 08	
57 49 41 33 25 17 09 01	
59 51 43 35 27 19 11 03	
61 53 45 37 29 21 13 05	
63 55 47 39 31 23 15 07	

1 mark each for
correct answer
→ 7 marks

4. As shown below, devices A and B can share a symmetric encryption key, K_s , by obtaining the key from a “key distribution center”, KDC, using an existing secure channel between A and KDC, and between B and KDC.

Marks
on Q4



- A. What purpose do messages numbered (4) and (5) serve?

Allows both initiator responder B to ascertain that initiator A has the session key K_s .

3 marks

- B. Can $f(N_2)$ in message (5) be replaced simply by N_2 ? Explain why you think this is true or false. In other words, briefly explain why the purpose identified by you in part A. above will be served or not if served if $f(N_2)$ in message (5) be replaced by N_2 .

False/True? NO. $f(N_2)$ in message (5) cannot be replaced simply by N_2 . This is so since any other device (say C) can simply reply with the received message (4), and claim it was able to decrypt message (4). This responder B cannot be sure whether it was A or some other device that responded.

4 marks